

Platform Engineering Technical Interview

Level: JL4 – Engineer

Sheet: JL4-2026-06-30-134819

Candidate name: Hristiyan Panoski

Interviewer: _____

Instructions:

- **Section A** (10 questions, 1 mark each): circle the letter of the correct answer.
- **Section B Part 1**: answer **both** questions in the space provided.

Section A – Multiple Choice

10 questions, 1 mark each. Circle the letter of the correct answer.

1. Looking at the Azure DevOps board, a new engineer sees her Tasks nested under larger(1 mark) items and wonders how her small pieces of work fit into the wider plan. In that hierarchy, your tasks connect upward to...
 - A. Only your laptop
 - B. Nothing
 - C. The payroll
 - D. **A User Story / Feature they help deliver - T**
2. On a tricky bug, a junior engineer is paired with a senior who shares her screen and talks(1 mark) through every decision out loud. Reflecting afterwards on why the team encourages this, pair-programming with a senior engineer mainly helps you...
 - A. Avoid work
 - B. Skip testing
 - C. **Learn faster and improve code quality - T**
 - D. Hide mistakes
3. Setting up branch policies, a lead configures the repository so that Pull Requests (PRs)(1 mark) can only complete with a rebase and fast-forward merge, rejecting the default merge-commit option. A new engineer, used to merge commits elsewhere, asks why the team is so particular about how branches land on main. Why does the team require a rebase & fast-forward merge?
 - A. It avoids running CI
 - B. It deletes the work item link
 - C. It allows skipping approvals
 - D. **It keeps a linear, traceable history - T**
4. An engineer pulls the next Task off the board and finds the requirement is vague - it is not(1 mark) obvious what 'done' would even look like, and the acceptance criteria are thin. Tempted to just start coding something plausible, they pause to consider the most professional move. You pick up a Task whose requirement is unclear. Best action?
 - A. Close it as done
 - B. Wait for the next status meeting
 - C. Guess and build something
 - D. **Clarify it against Definition of Ready before starting -T**

5. A new engineer joins the Platform Engineering team at a chip foundry and reads the(1 mark) coding standards on day one, where every reviewer keeps stressing 'clean, testable code.' During onboarding, their mentor asks them to pick the option that captures that team value. Which best describes the clean, testable code the team expects?
- A. Defers all tests until release
 - B. Mixes I/O and business logic for speed
 - C. **Small, single-responsibility units with logic separated from side effects - T**
 - D. Relies on global mutable state
6. A team refactoring a wafer-data parser adopts the Functional Core / Imperative Shell(1 mark) (FC/IS) pattern, reorganizing the module into two distinct layers. A junior engineer wiring up the new structure wants to be sure exactly what belongs in the inner layer before moving code around. In the FC/IS pattern, the 'core' should be...
- A. The user interface
 - B. The CI pipeline
 - C. Where file and network I/O happen
 - D. **Pure functions with no side effects - T**
7. While tracing how their new rule compiler reads a file, an engineer keeps seeing a first(1 mark) stage labelled the lexer, or tokenizer, that runs before anything else. Curious about what that stage actually does, she checks her understanding of the basics. A lexer/tokenizer...
- A. Executes the program
 - B. Draws the AST
 - C. Deploys code
 - D. **Breaks input text into tokens - T**
8. A senior walks a new hire through the rule compiler and shows two tree structures built(1 mark) from the same input: a parse tree, also called a concrete syntax tree, produced directly by the parser, and a separate Abstract Syntax Tree (AST) consumed by later stages. The new hire wonders why the tool bothers keeping two different trees. How does an AST differ from a parse (concrete syntax) tree?
- A. They are identical
 - B. The AST stores characters
 - C. **The AST omits incidental syntax (punctuation) and keeps meaning - T**
 - D. The AST has more nodes always
9. Reviewing a layout, a checker flags an enclosure rule on a structure where two stacked(1 mark) layers do not relate the way the deck demands. The layout engineer, comfortable with spacing and width rules, has rarely dealt with this third family of constraints that ties one layer to another. Her mentor asks her to state precisely what such a rule demands. An enclosure rule requires...

- A. **One layer to surround another by at least a minimum amount (e.g., metal over contact) - T**
 - B. Shapes to be deleted
 - C. Two shapes to overlap exactly
 - D. Layers to be merged
10. A reviewer points an engineer at pytest's parametrize feature after seeing a cluster of(1 mark)
very similar tests, and the engineer, new to it, asks what parametrize is actually meant
for. pytest's parametrize is used to...
- A. Mock the database
 - B. **Run the same test over multiple inputs - T**
 - C. Generate coverage
 - D. Deploy to many environments

Section B – Written Questions

Support each answer with a worked example or codebase reference – unsupported answers earn half marks at most.

Part 1 – Answer BOTH questions.

1. During a new-node kickoff, a customer engineer repeatedly hears the foundry team refer to the technology's 'ground rules' as the reference point for the whole design effort. Curious what specifically sits behind that label, he asks a process contact to define the term. What are ground rules?

Ground rules are set of rules that are defined by physical limitations and constraints of a manufacturing process. They are provided by the foundry, as a part of a PDK – Process Design Kit. Depending on the manufacturing processes, types of implantations, etching, light wavelength (used for photolithography), manufacturing machines and other... The foundry sets ground rules to ensure high quality, high yield, and long-lasting chips.

Ground rules include design rules (minimal size of details, minimal spacing), electrical and reliability constraints (electromigration, electrical discharges) and other rules and constraints.

An example of ground rules – design rules. If there are two parallel wires (Net1 and Net2) on Metal 1, one of which (Net1) ends in a via up to Metal 2, and the other (Net2) continues on Metal 1. The design rules would check if:

- Net1 and Net2 wires have at least the minimal width for Metal 1.

- Net1 and Net2 wires are spaced at least with minimal Metal 1 spacing (for separate features).

- Where Net1 has via up to Metal 2, is Via1 layer enclosed by minimal enclosure of Metal 1.

- Is Via1 enclosed by minimal enclosure of Metal 2.

- Dose Net1 on Metal 2, have minimal width for Metal 2, and dose it have minimal area.

As well as other checks to ensure correct manufacturing.

https://github.com/PanoEth/Interview_GF/tree/main/SectionB_q1

2. An engineer at the foundry has just pushed what they believe is a finished Task, but the Continuous Integration (CI) run comes back red with two failing tests. A reviewer is available right now and it is tempting to hand it over and sort the failures out in parallel. Before doing anything rash, the engineer thinks through the proper sequence. You finished coding a Task but tests are failing in CI. Walk through what you do before requesting review.

Before sending task for a review, the engineer must ensure that the CI passes. First, the engineer should check the CI output to identify what causes the failure. Next, they should recreate failing cases locally on PC/Laptop. Since the CI, does more than just ensure that the code is safe to compile, the engineer should pay attention to what the root cause is. If the fails occur during randomized automated tests, it typically means that the problem is isolated. It could be caused by input reading error, edge case error, large data/resource requirement, logic error, output format etc. After finding out the root cause, the code should be updated (debugged), making sure that the cases won't fail (locally). Additional testing with similar data is good practice. After ensuring that the errors don't occur he should push the updated code and wait for a pass of the CI. **Only after making sure that the CI is passed, should he request a review.**

https://github.com/PanoEth/Interview_GF/tree/main/SectionB_q2